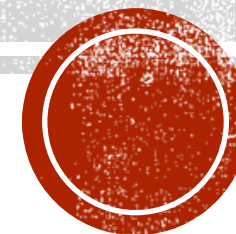


Recovering Germanium from waste Optical fiber cable



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The Formosa Association of Resource Recycling—Secretary-general

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2014 November

Outline



- **Circumstances of Germanium around the world**
- **Application of Germanium**
- **Recovery of germanium from fiber-optic cables**
- **Summaries**



Introduction of germanium



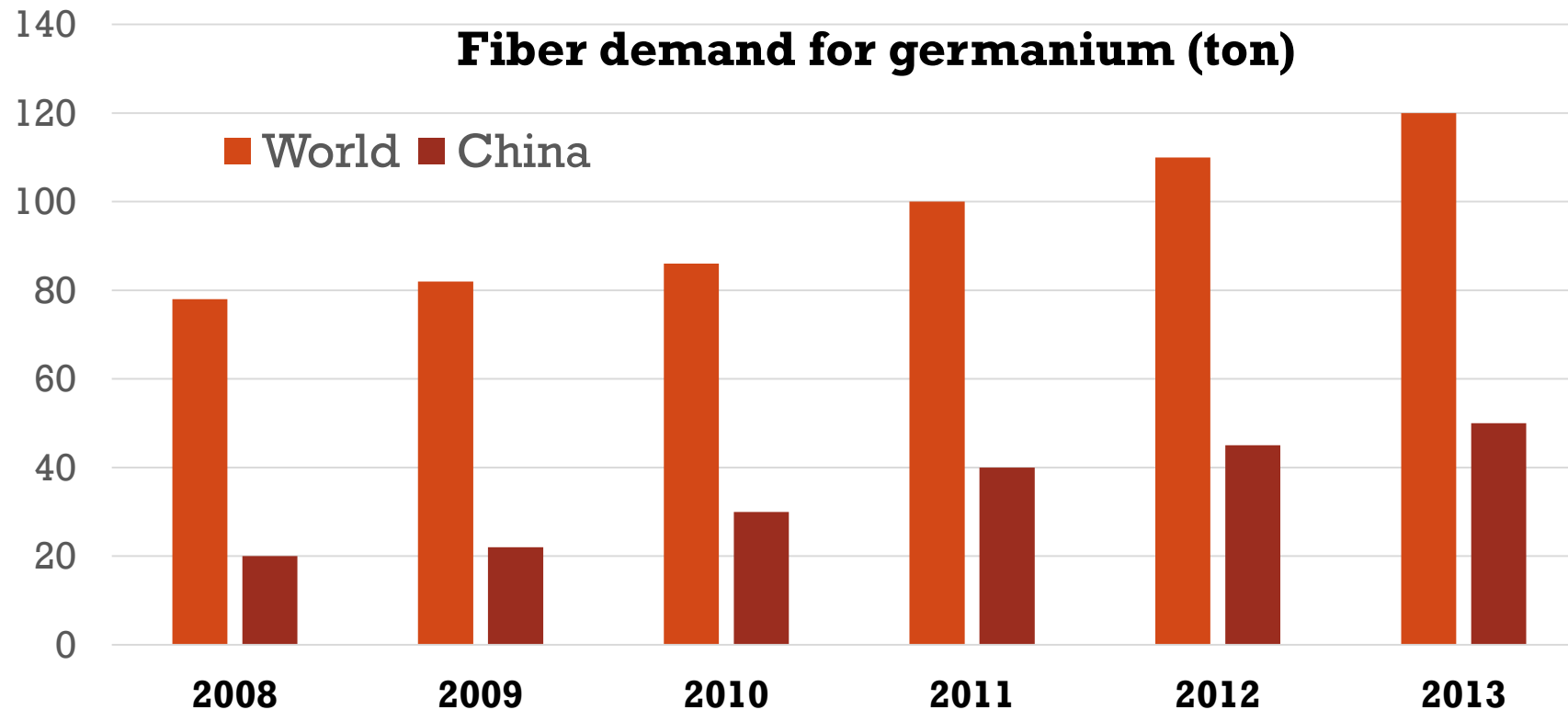
- Germanium(Ge) : **Rare metal**(Scattered Metals)
- Melting Point : 937.4°C
- Boiling Point : 2830°C
- Density : 5.35g/cm^3
- Hardness : $6\sim 6.5$
- Germanium has a **great semiconductor property** and is an important material of the semiconductor.
- Arrangement of atoms in crystals of germanium is same as diamond , its **chemical characteristic is stable.**
- Insoluble in : **dilute acids and alkalis**
- Dissolves slowly in : **concentrated sulfuric acid** and reacts violently with **molten alkalis** to produce germanates ($[\text{GeO}_3]^{2-}$)



Reserves and Distribution



- The reserves of Germanium : **8,600 tons**
- The **average annual demand** of growth rate is about **33.89 %** .
- Current world's reserves of germanium is only enough for **40 years**.



Reserves and Distribution



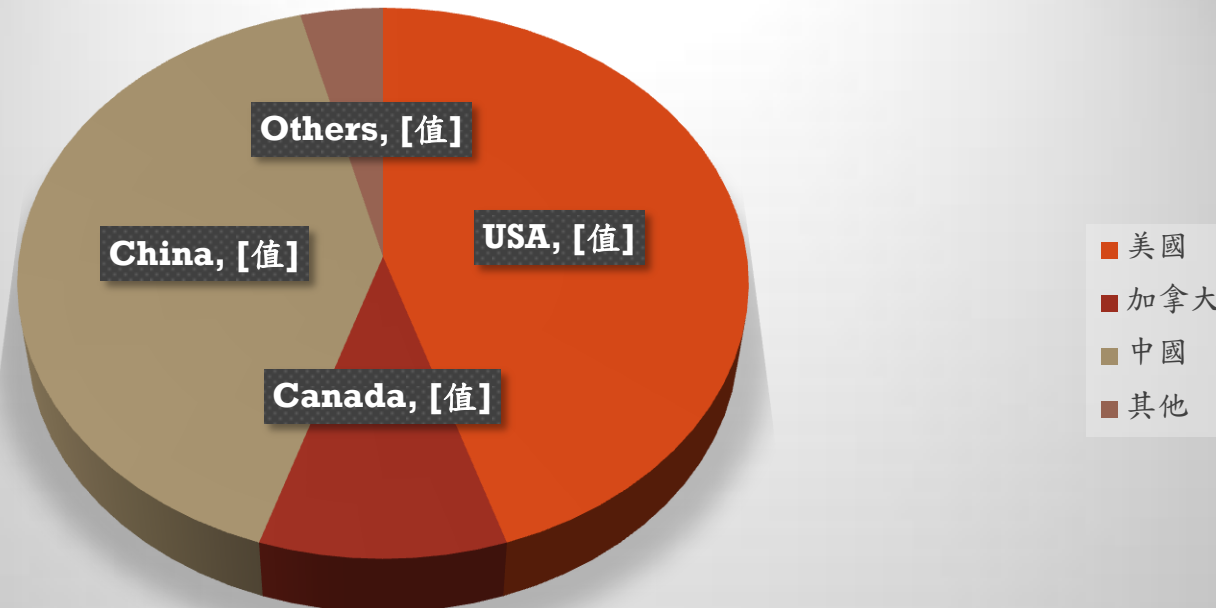
The largest country in production :

China

72 percent of global production of germanium

The production of germanium in USA is subject to the production of **lead** and **zinc**.

US annual production of germanium : **4.6 tons**,
(Global production : **3%**)



Distribution of germanium reserves

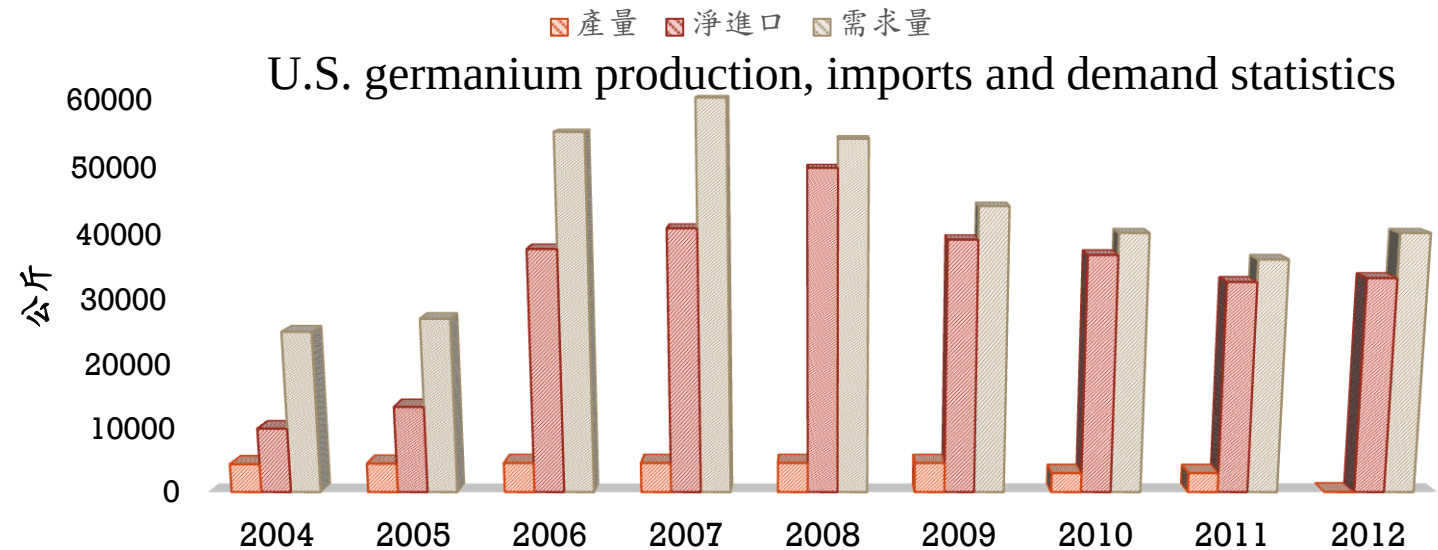


Reserves and Distribution



- Early in 1984 , the United States made **germanium** as a defense germanium reserves of **strategic resources** protection.
- U.S Annual consumption : **60 tons** , Imports accounted for : **86.67%**.

Imported Source:
China : 51% ,
 Belgium 24%,
 Russia : 16%;
 Germany:6%, other 3%.



	2004	2005	2006	2007	2008	2009	2010	2011	2012
production → 產量	4400	4500	4600	4600	4600	4600	3000	3000	0
imports → 淨進口	10000	13400	37600	40700	49700	39000	36700	32600	33200
demand → 需求量	25000	27000	55000	60000	54000	44000	40000	36000	40000

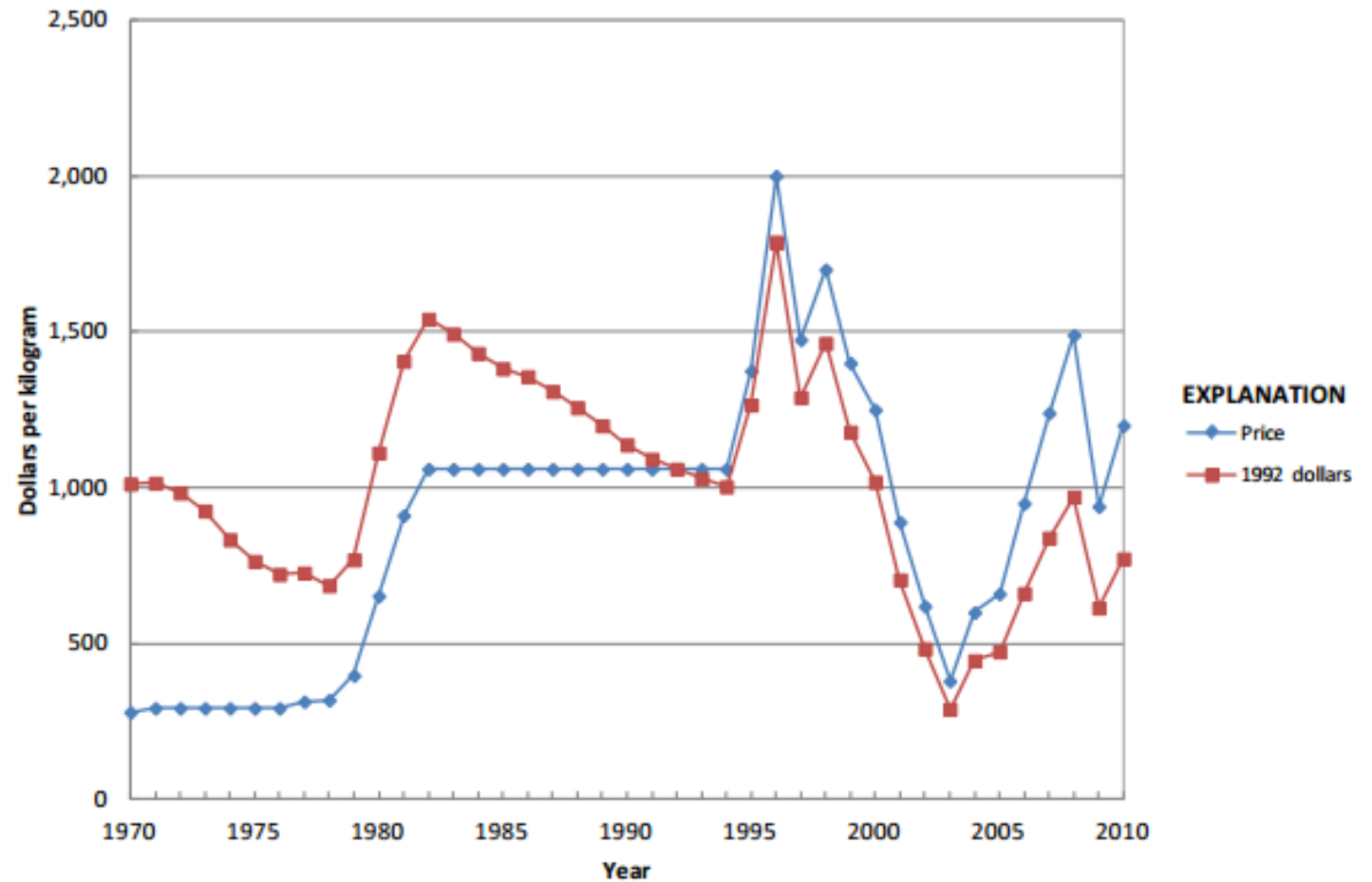
Source : USGS year statistics



Market of Germanium



5N Germanium ingot	RMB/Kg
2014e	12,000
2013	11,625
2012	11,250
2011	10,875
2010	10,500
2009	10,125
2008	9,750
2007	9,375



■ Germanium dioxide 99.99%

\$ 1,435 / kg



Market of Germanium



Tables: import and export statistics germanium Taiwan

Ge	Import			Export		
	Kg	Total price	Unit price (kg/dollae)	Kg	Total price	Unit price (kg/dollar)
98	782	40,170,000	51,368	131	5,835,000	44,542
99	4,011	133,723,000	33,339	658	23,858,000	36,258
100	8,503	234,817,000	27,616	77	5,104,000	66,286
101	10,672	304,193,000	28,504	274	5,499,000	20,069
102	10,625	371,781,000	34,991	334	12,987,000	38,883
Source：財政部關稅總局統計資料						



The Source Of Germanium-Mine

- Germanium content in the crust : **0.0007%** (Seven millionths)
- Dispersed exist in other minerals:

鍺石 Germanium stone (including germanium 10%),

硫鍺鐵銅礦 Copper iron sulfide germanium (Ge containing 7.7%),

硫銀鍺礦 Argyrodite (germanium-containing 6.7%),

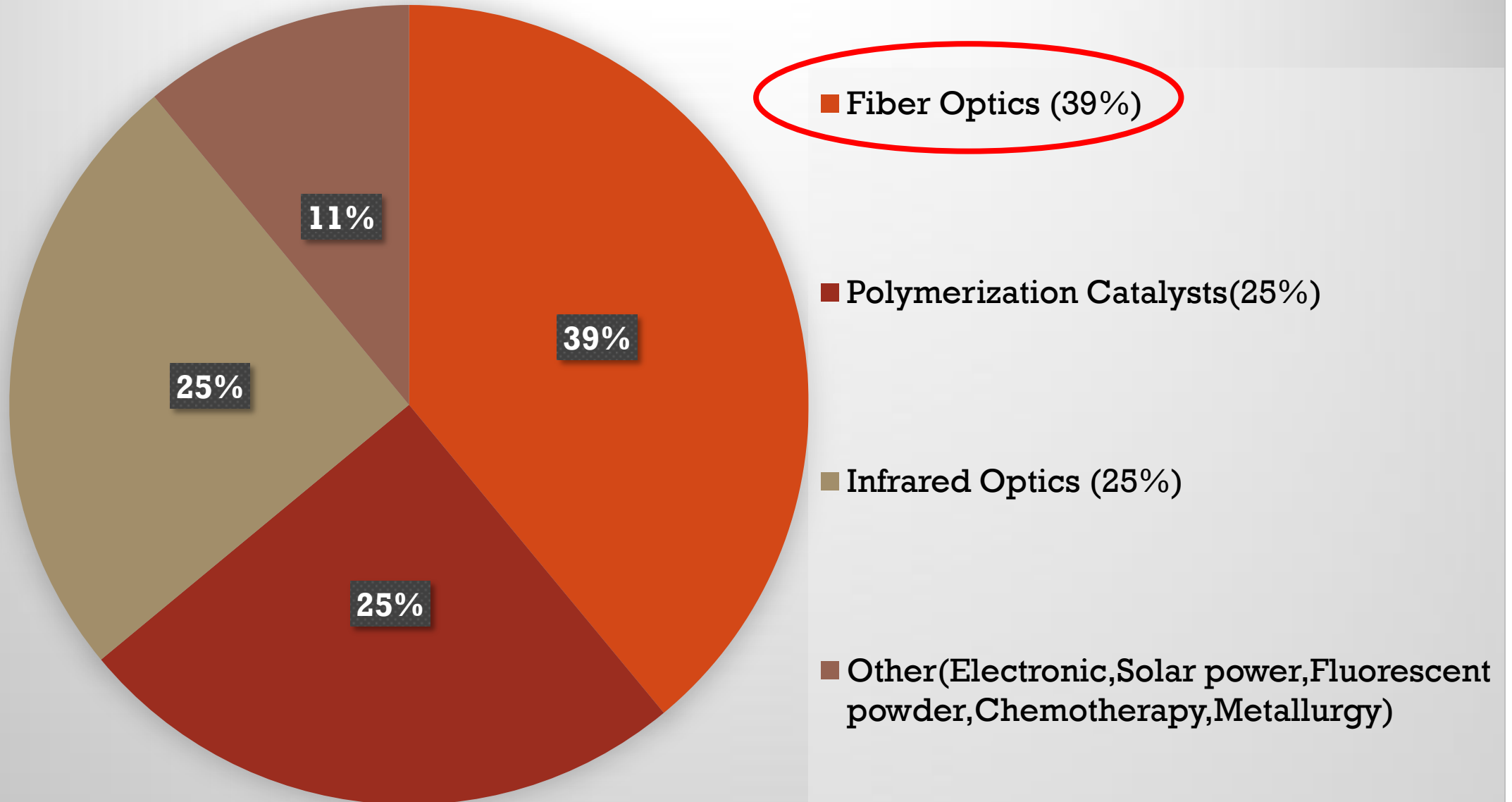
黑硫銀錫礦 Black silver tin sulfide (germanium-containing 1.8%),

mainly in industry recycled or recovered as a byproduct from coking Dust while processing sulfide ore.

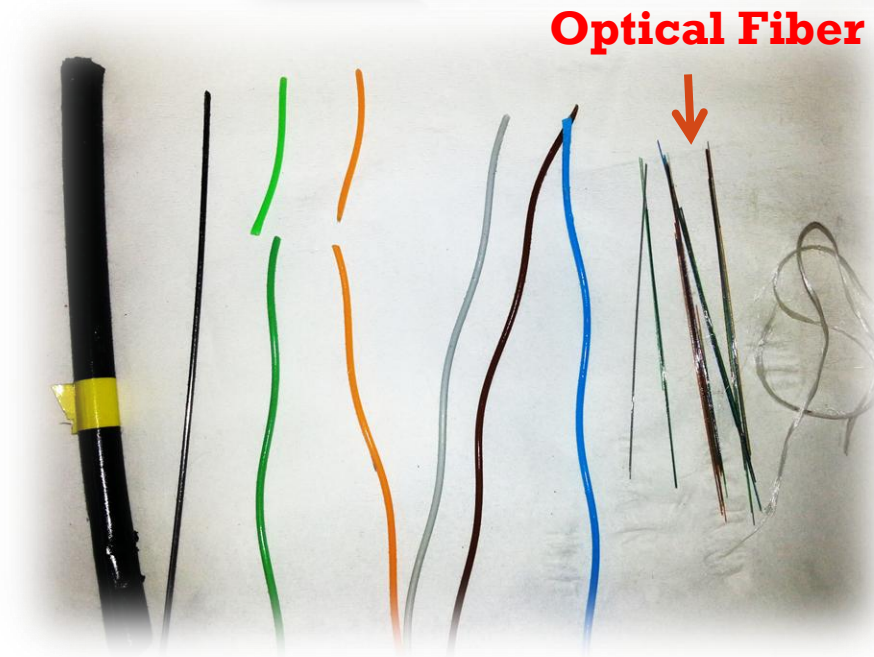
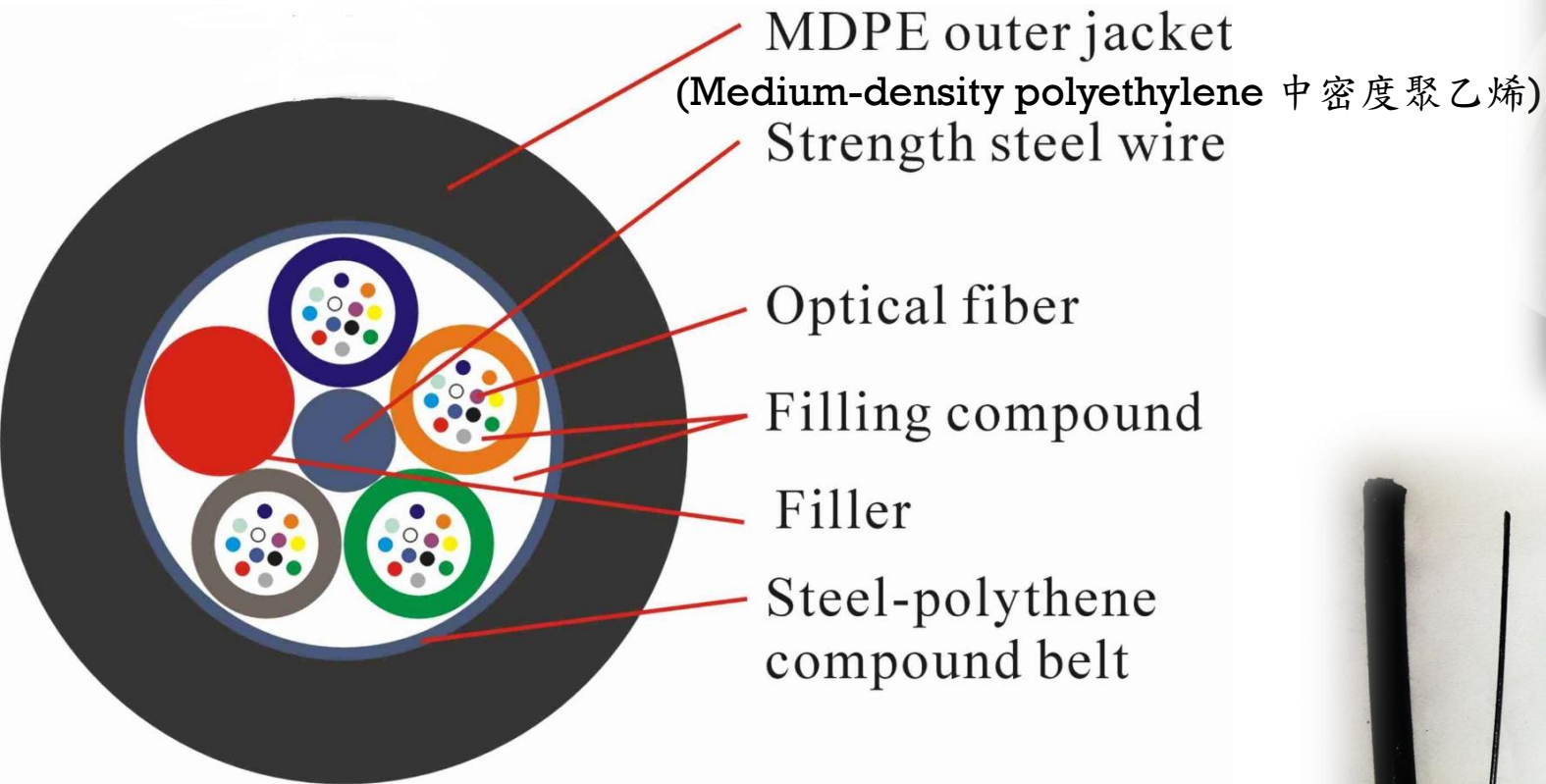
- Germanium ore containing more than **20ppm** has recovery value.



Germanium World Consumption Structure in 2013



Optical fiber cable



Optical fiber cable



Proportion by Weight

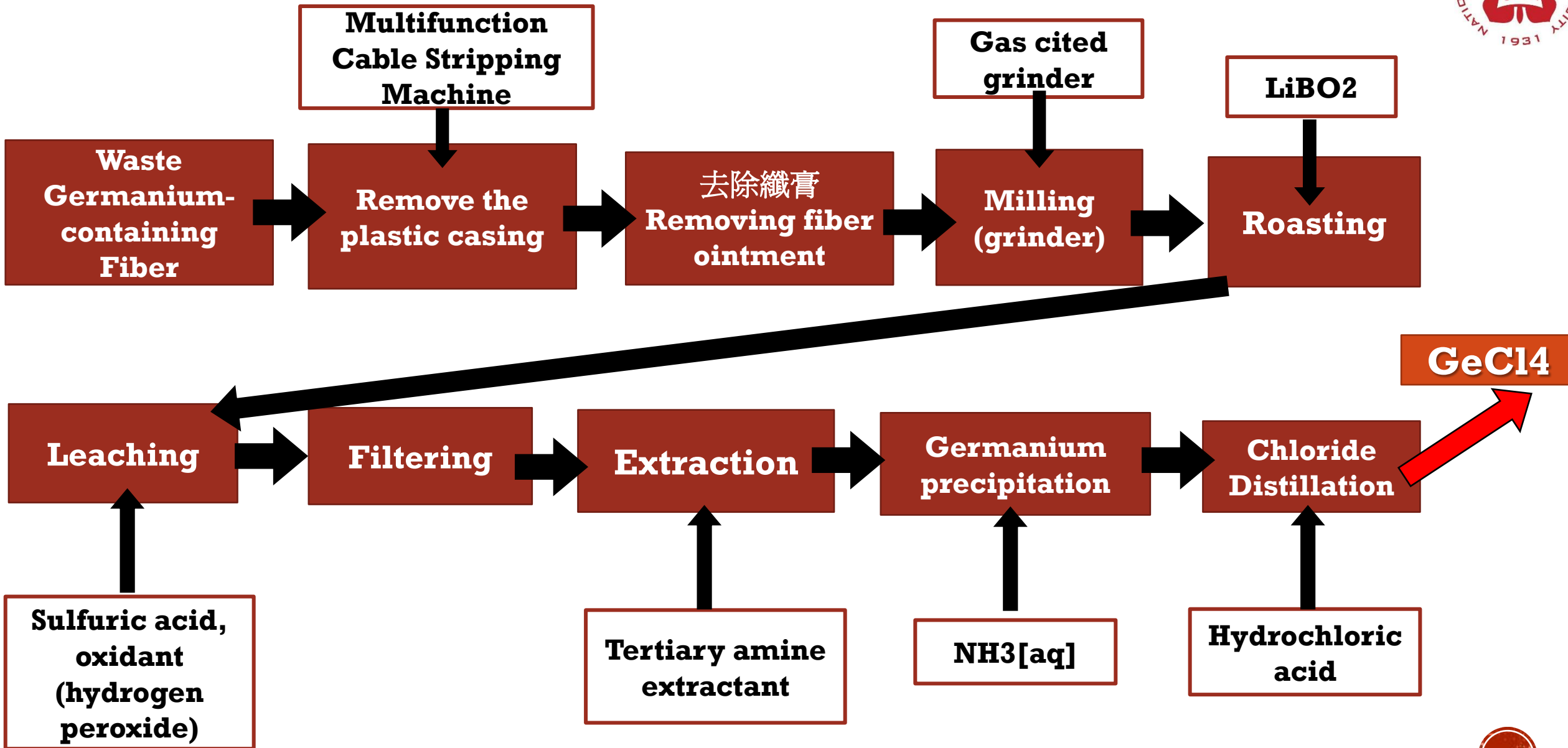
Outer Jacket (MDPE)	Optical Fiber	Steel wire	Filling compound (Thin plastic wire 、 fiber ointment.etc)
60%	10~15%	10~15%	10~15%

Germanium optical fiber element content analysis

Si	Ge	GeCl ₄
95.17%	3%	10%



Recycling Process of Optical fiber



Remove the plastic casing- Multifunction Cable Stripping Machine



Voltage power:

220V / 2.2KW (Motor: 3HP)

Cutting capacity: 1mm ~ 40mm,
a total of 20 stringing holes

Scope: computer line to 325 square
feet of wire (cable) line

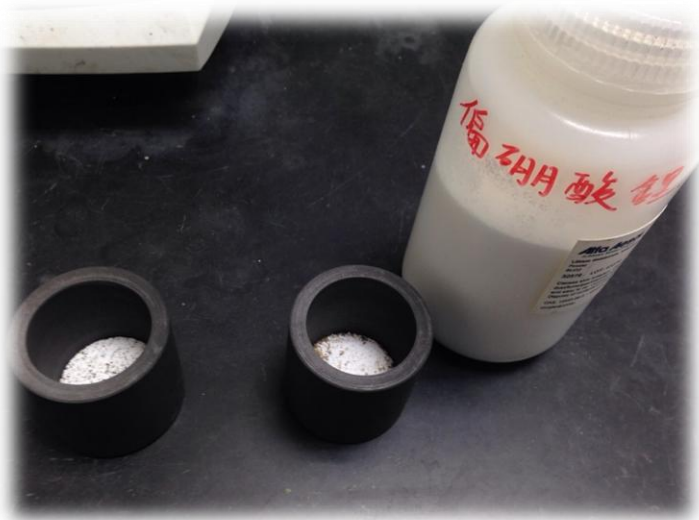
Milling (grinder) -Gas cited grinder

Milling to at least 200 mesh



Roasting

- (1) Weighed Germanium-containing fiber abrasive in platinum crucibles (graphite crucible), add toner and flux (**Lithium metaborate LiBO_2**) mixed. (Fiber abrasive / flux = $1/5 \sim 1/10$)
- (2) Placed in a furnace heated to $800 \sim 1000^\circ\text{C}$.
- (3) heating from 10 to 20 minutes, removed and quickly placed into cold water to cool down, then condensed as a solid balls.



Leaching

- 0.1 ~ 1M sulfuric acid was added (S / L = 1 / 1.5 ~ 2), adding hydrogen peroxide as the oxidant (S / L = 1 / 0.2 ~ 1).
- The reaction at a temperature of 2 to 2.5 hours at 80 ~ 100 ° C condition.



Solvent extraction

- (1) Selection of **Alamine 308** Trioctylamine (**R3N**): extraction agent 30%, 15% modifier (TBP), diluent (kerosene) 55% as the organic extraction system.
- (2) The aqueous phase acidity pH: 2.0 ~ 2.5.
- (3) The volume of the aqueous phase and an organic phase ratio
 $O / A = 1/3 \sim 4$
- (4) Extraction of the reaction time from 5 to 10 minutes.
- (5) Ambient temperature.
- (6) Germanium extraction rate was more than **97%**



Germanium precipitation/ Chloride Distillation



- (1) Adding Ammonium hydroxide ($\text{NH}_3(\text{aq})$) for Germanium precipitation.
- (2) Add 9 mol / l hydrochloric acid to dissolve, S/L 1: 1 ~ 3, the reaction at $100-110^\circ\text{C}$, 2-4 hours to obtain steam at a temperature of $50-70^\circ\text{C}$ is condensed, collecting the liquid and will get Germanium tetrachloride (GeCl_4).
- (3) $\text{Ge}(\text{OH})_4 + 4\text{HCl} \rightarrow \text{GeCl}_4 + 2\text{H}_2\text{O}$ ◦
- (4) germanium recovery rate was above **98%**.



**THANKS FOR
YOUR ATTENTION**

